Phys. 2110 Solution

HW 02 : 2.4, 19, 30, 31, 38, 50, 51, 95

2.4 (a) The eastward run takes time $\frac{200\,m}{5.0\,m/s} = 40.0\,s$

The westward run takes time $\frac{280\,m}{4.0\,m/s} = 70.0\,s$

the average speed is $\frac{480\,m}{110.0\,s} = \boxed{4.4\,m/s}$

(b) the average velocity $V_{av\text{-}x} = \frac{\Delta x}{\Delta t} = \frac{-80\,m}{110.0\,s} = \boxed{-0.73\,m/s}$

the average velocity is directed westward.

2.19 (a) V_x is the slope of the x versus t curve and a_x is the slope of the V_x versus t curve

$t = 0 \sim 5\,s$: x versus t is a parabola, so a_x is a constant. The curvature is positive so a_x is positive.

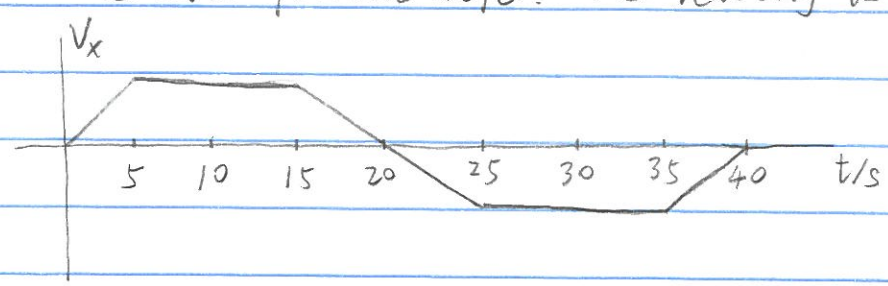
V_x versus t is a straight line with positive slope, $V_{0x} = 0$

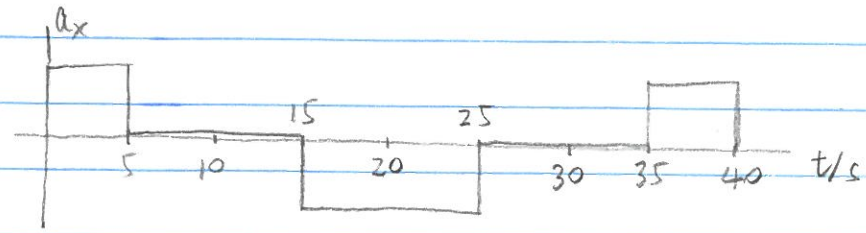
$t = 5 - 15\,s$: x versus t is a straight line with positive slope so V_x is constant and positive. $a_x = 0$

$t = 15 - 25\,s$: x versus t is a parabola with negative curvature, so a_x is constant and negative. V_x versus t is a straight line with negative slope. The velocity is zero at 20s.

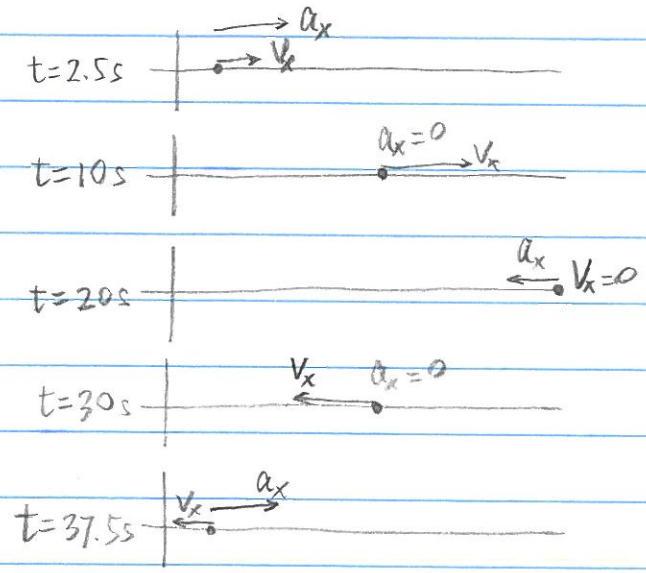
$t = 25 - 35\,s$: x versus t is a straight line with negative slope. so V_x is constant and negative, $a_x = 0$

$t = 35 - 40\,s$: x versus t is a parabola with positive curvature, so a_x is constant and positive. V_x versus t is a straight line with positive slope. The velocity is zero at 40s.





(b) The motion diagrams are sketched below



2.30 (a) For $t = 0 - 8s$: $x = \left(\frac{V_{0x} + V_x}{2}\right)t = \left(\frac{0 + 20m/s}{2}\right) \times (8s) = 80\,m$

$$a_x = \frac{V_x - V_{0x}}{t} = \frac{20\,m/s}{8\,s} = 2.50\,m/s^2$$

Constant speed for 60m: The car moves from $x = 80m$ to $x = 140\,m$, $V_x = 20\,m/s$ $a_x = 0$

This interval is $\Delta t = \frac{60\,m}{20\,m/s} = 3\,s$ $t = 8s + 3s = 11s$

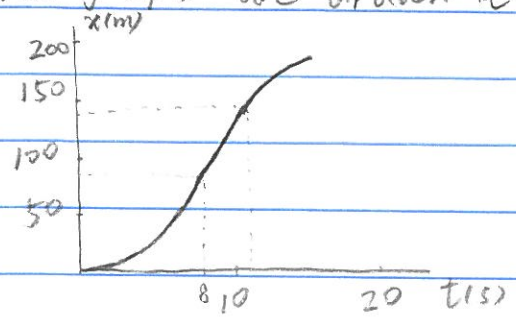
Slowing from 20m/s until stopped: The car moves from $x = 140m$ to $x = 180m$. The velocity decreases linearly from 20m/s to 0.

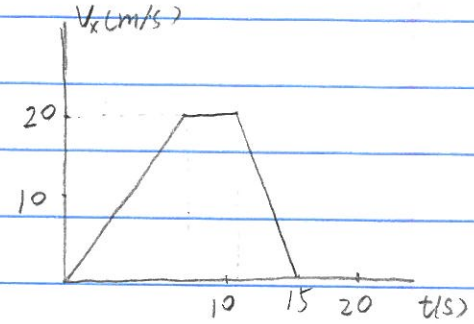
$$x - x_0 = \left(\frac{V_{0x} + V_x}{2}\right)t \Rightarrow t = \frac{2 \times 40m}{20m/s + 0} = 4s$$

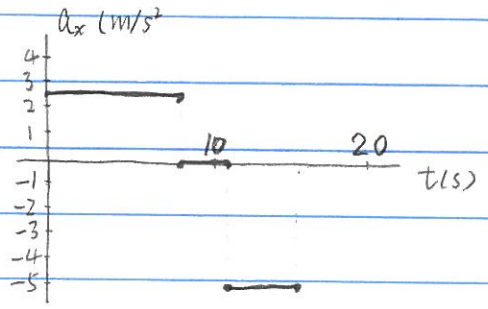
This segment is from $t = 11s$ to $t = 15s$

$$V_x^2 = V_{0x}^2 + 2a_x(x - x_0) \Rightarrow a_x = \frac{-(20.0\,m/s)^2}{2 \times 40\,m} = -5.00\,m/s^2$$

The graphs are drawn below

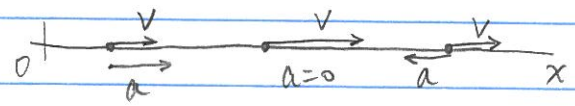






(a)

(b) The motion diagram is sketched below



2.31 (a) At $t = 3s$, the V_x versus t is a horizontal straight line, with zero slope. Thus $\boxed{a_x = 0}$

At $t = 7s$, the V_x versus t is a straight line with slope

$$\frac{45\,m/s - 20\,m/s}{9s - 5s} = 6.3\,m/s^2, \text{ thus } \boxed{a_x = 6.3\,m/s^2}$$

At $t = 11s$, $a_x = \dfrac{0 - 45\,m/s}{13s - 9s} = \boxed{-11.2\,m/s^2}$

(b) During the first 5 second the acceleration is constant,

$V_{0x} = 20\,m/s$, $a_x = 0$, $t = 5s$. $x - x_0 = ?$

$x - x_0 = V_{0x}t$ (no $\frac{1}{2}a_x t^2$ term since $a_x = 0$)

$x - x_0 = 20\,m/s \times 5s = \boxed{100\,m}$, this is the distance the officer goes in the first 5 seconds

During the interval $t = 5s$ to $9s$, the acceleration is constant $a_x = 6.25\,m/s^2$, the interval $\Delta t = 4s$

$V_{0x} = 20\,m/s$, $x_0 = 100\,m$, $x - x_0 = ?$

$x - x_0 = V_{0x}t + \frac{1}{2}a_x t^2 = V_{0x}\Delta t + \frac{1}{2}a_x(\Delta t)^2$

$x - x_0 = 20\,m/s \times 4s + \frac{1}{2} \times 6.25\,m/s^2 \times (4s)^2 = 130\,m$

$x = x_0 + 130\,m = 100\,m + 130\,m = \boxed{230\,m}$, this is the distance the officer goes in the first 9 seconds.

During the interval $t = 9s$ to $13s$, the acceleration is again constant, $a_x = -11.2\,m/s^2$, the interval $\Delta t = 4s$

$V_{0x} = 45\,m/s$, $x_0 = 230\,m$, $x - x_0 = ?$

$x - x_0 = V_{0x}t + \frac{1}{2}a_x t^2 = V_{0x}\Delta t + \frac{1}{2}a_x(\Delta t)^2$

$x - x_0 = 45\,m/s \times 4s + \frac{1}{2} \times (-11.2\,m/s^2) \times (4s)^2 = 90.4\,m$

$x = x_0 + 90.4\,m = \boxed{320\,m}$

so the officer goes 320 m in the first 13 seconds.

2.38 (a) Assume an initial height of 200m,

$y - y_0 = 200\,m$, $a_y = 9.80\,m/s^2$, $V_{0y} = 0$, $V_y^2 = V_{0y}^2 + 2a_y(y - y_0)$

$V_y = \sqrt{2 \times 9.80\,m/s^2 \times 200\,m} = 60\,m/s = 200\,km/h = 140\,mi/h$

(b) The speed with which raindrops actually strike the ground is about 1 m/s.

(c) The actual speed at the ground is much less than the speed calculated, so neglect of air resistance is not a good approximation for falling raindrops.

2.50 (a) $V_x = V_{0x} + \int_0^t 2t\,dt = V_{0x} + \frac{1}{2}2t^2 = V_{0x} + (0.60\,m/s^2)\,t^2$

$t = 1.0\,s,\quad V_x = 5.0\,m/s \quad \Rightarrow \quad V_{0x} = 4.4\,m/s$

Then $t = 2.0\,s \qquad V_x = 4.4\,m/s + (0.60\,m^2/s^3)\times(2.0\,s)^2 = \boxed{6.8\,m/s}$

(b) $x = x_0 + \int_0^t (V_{0x} + \frac{1}{2}2t^2)\,dt = x_0 + V_{0x}t + \frac{1}{6}2t^3$

$= x_0 + (4.4\,m/s)t + (0.2\,m/s^3)\,t^3$

$t = 1.0\,s,\quad x = 6.0\,m \quad \Rightarrow \quad x_0 = 1.4\,m$

Then $t = 2.0\,s,\quad x = 1.4\,m + (4.4\,m/s)\times 2.0\,s + (0.2\,m/s^3)\times(2.0\,s)^3$

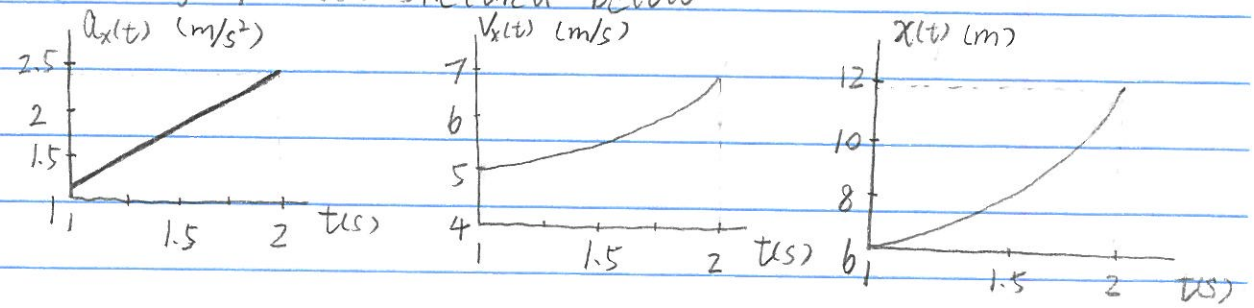
$= \boxed{11.8\,m}$

(c) $a_x(t) = (1.20\,m/s^3)\,t$

$V_x(t) = 4.4\,m/s + (0.60\,m/s^3)\,t^2$

$x(t) = 1.4\,m + (4.4\,m/s)\,t + (0.20\,m/s^3)\,t^2$

The graphs are sketched below



2.51 (a) $V_x = V_{0x} + \int_0^t a_x\,dt = V_{0x} + \int_0^t (At - Bt^2)\,dt = V_{0x} + \frac{1}{2}At^2 - \frac{1}{3}Bt^3$

At rest at time $t=0$ means $V_{0x} = 0$

$V_x = \frac{1}{2}\times(1.50\,m/s^3)\,t^2 - \frac{1}{3}(0.120\,m/s^4)\,t^3$

$= \boxed{(0.75\,m/s^3)\,t^2 - (0.040\,m/s^4)\,t^3}$

$x = x_0 + \int_0^t V_x\,dt = x_0 + \int_0^t (\frac{1}{2}At^2 - \frac{1}{3}Bt^3)\,dt$

$= x_0 + \frac{1}{6}At^3 - \frac{1}{12}Bt^4$

At the origin at $t=0$ means $x_0 = 0$

5 $x = \frac{1}{6}At^3 - \frac{1}{12}Bt^4 = \boxed{(0.25\,m/s^3)\,t^3 - (0.010\,m/s^4)\,t^4}$

(b) $a_x = \frac{dV_x}{dt} = 0$ so $At - Bt^2 = 0$ $\quad t(A - Bt) = 0$

One root is $t=0$, but at $t=0$, $V_x = 0$ and not a maximum

The other root is $t = \frac{A}{B} = \frac{1.50\ m/s^3}{0.120\ m/s^4} = 12.5\ s$

At this time $V_x = (0.75\ m/s^3)\ t^2 - (0.040\ m/s^4)\ t^3$ gives

$V_x = (0.75\ m/s^3) \times (12.5\ s)^2 - (0.040\ m/s^4) \times (12.5\ s)^3$

$= 117.2\ m/s - 78.1\ m/s = \boxed{39.1\ m/s}$

2.95 Let the students' speed be V_0 and the bus's initial position be x_0. The initial position of the student is taken to be zero, and the initial velocity of the bus is taken to be zero.

The positions of the student x_1 and the bus x_2 as functions of time are then $x_1 = V_0 t$ and $x_2 = x_0 + \frac{1}{2} a t^2$

(a) $x_1 = x_2$, solve for the time t gives $t = \frac{1}{a}(V_0 \pm \sqrt{V_0^2 - 2ax_0})$.

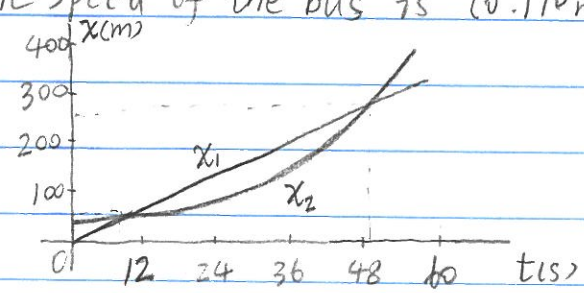
$t = \frac{1}{0.170\ m/s^2}\left(5.0\ m/s \pm \sqrt{(5.0\ m/s)^2 - 2 \times (0.170\ m/s^2) \times (40.0\ m)}\right)$

$= 9.55\ s$ and $49.3\ s$

The student will catch the bus at $t = \boxed{9.55\ s}$, at this time the student has run a distance $V_0 t = (5.0\ m/s) \times 9.55\ s = \boxed{47.8\ m}$

(b) The speed of the bus is $(0.170\ m/s^2) \times (9.55\ s) = \boxed{1.62\ m/s}$

(c)



(d) At the later time, the student has passed the bus, maintaining her constant speed, then the accelerating bus catches up to her.

At this later time the bus's velocity is

$(0.170\ m/s^2) \times (49.3\ s) = \boxed{8.38\ m/s}$

(e) $\boxed{\text{NO}}$; when the student runs at $V_0 = 3.5\ m/s$

$V_0^2 = (3.5\ m/s)^2 < 2 \times (0.170\ m/s^2) \times (40.0\ m) = 2aX_0$

the roots of the quadratic are imaginary.

(f) For the student to catch the bus, $V_0^2 > 2aX_0$, so the minimum speed is

$\sqrt{2 \times (0.170\ m/s^2) \times (40.0\ m)} = \boxed{3.688\ m/s}$.

She would run for a time $\frac{3.688\ m/s}{0.170\ m/s^2} = \boxed{21.7\ s}$, and the distance $(3.688\ m/s) \times (21.7\ s) = \boxed{80.0\ m}$